

SIMARPREET KAUR

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PROFESSIONAL SUMMARY

Computer Science graduate who builds end-to-end machine learning and NLP systems — from data pipelines and model training through evaluation and deployment. Experience spans transformer fine-tuning, multilingual semantic search, and real-time speech-to-structured-data pipelines in Python, PyTorch, and TensorFlow. Concurrently leads retail operations with P&L ownership, bringing firsthand operational problem-solving to technical work.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, HTML/CSS

ML & Deep Learning: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, Transformers (BERT, DistilBERT, DeBERTa), Sentence-Transformers (SBERT), NLTK

NLP & Data: Semantic search, embeddings, text classification, summarization, information extraction, speech-to-text (Whisper); Pandas, NumPy, Matplotlib, Seaborn, Tableau, Power BI

Engineering & Methods: Git/GitHub, Flask, REST APIs, Azure ML, model deployment; feature engineering, cross-validation, hyperparameter tuning, class imbalance (SMOTE), error analysis, F1 / ROC-AUC / precision / recall

PROJECTS

Pocket AI — Real-Time Speech to Structured Data TensorFlow, Whisper, NLP

- Trained a Conv1D + GRU wake-word model in TensorFlow/Keras on a custom audio dataset, handling severe positive-frame class imbalance via weighted loss and threshold tuning.
- Integrated Whisper for speech-to-text and built a parsing engine converting 90% of free-form transcripts into structured JSON capturing task IDs, actions, timestamps, and assignees.
- Converted the trained model to TFLite and tuned the real-time listener — frame-level inference, debouncing, and EMA smoothing — for low-latency on-device detection.

Sarcasm Detection — Transformer Fine-Tuning DistilBERT, DeBERTa, PyTorch

- Fine-tuned and benchmarked DistilBERT and DeBERTa on a 202K-comment Reddit corpus, incorporating parent-comment context as an input feature.
- DeBERTa reached 74.1% accuracy and 0.750 macro F1 against a 72.9% / 0.728 DistilBERT baseline; used macro-averaging to account for class balance.
- Authored the annotation guidelines defining label boundaries for an inherently subjective classification task.

Multilingual News Explorer — Cross-Lingual Semantic Search SBERT, Flask, PyTorch

- Built semantic retrieval across 8 languages with multilingual SBERT embeddings, plus extractive summarization via similarity graphs achieving 80% text reduction.
- Deployed a PyTorch-backed Flask inference API with precomputed embeddings, serving queries in under 200ms.
- Resolved production deployment issues including dependency conflicts, NLTK corpus loading, and NumPy version incompatibilities.

WORK EXPERIENCE

Team Lead, Operations — Best Buy Oct 2022 – Present

Progression: Specialist → Senior Specialist → Team Lead, Operations

- Review P&L and key business metrics to assess performance, identify opportunities, and direct operational priorities.
- Investigate inventory discrepancies and operational exceptions to root cause, implementing process changes that improve stock accuracy.
- Lead hiring, onboarding, training, coaching, and performance management across the operations team.
- Own daily operations across inventory, fulfillment, merchandising, and customer service; identified a stock-tracking gap and built, deployed, and iterated an internal tool now in daily use by staff.

Data Science Analyst — Outlier Oct 2025 – Apr 2026

- Designed structured evaluation inputs for LLM benchmarking, assessing output quality, correctness, and analytical rigor.
- Performed error analysis to identify recurring failure patterns, applying F1, ROC-AUC, and precision/recall to benchmark performance.

EDUCATION

McMaster University — Honours Bachelor of Computer Science Graduated May 2026

Relevant coursework: Machine Learning, Natural Language Processing, Databases, Operating Systems

St. Lawrence College — Computer Networking and Technical Support Sep 2017 – Jun 2019